

9(a)(i)	discrete/ packet / quantum of energy of electromagnetic radiation	[1] [1]
(a)(ii)	wavelength = $(6.63 \times 10^{-34}) / (9.5 \times 10^{-28}) = 700 \times 10^{-9} \text{ m}$ so red	[1] [1]
(a)(iii)	pressure = force / area force = rate of change of momentum $= 2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}$ pressure = $(2 \times 9.5 \times 10^{-28} \times 1.4 \times 10^{15}) / (2.5 \times 10^{-6})$ $= 1.1 \times 10^{-6} \text{ Pa}$	[1] [1]
(b)(i)	greater photon energy (hf) and same work function (Φ) so by $hf = \Phi + KE_{\text{max}}$ maximum kinetic energy is increased	[1] [1]
(b)(ii)	By $I = Nh\nu/At$ or $P = N(h\nu)/t$, greater photon energy (and same power) so lower number of photons (per unit time) each electron absorbs/interacts with one photon so lower rate of emission	[1] [1]
(c)(i)	(Electron) diffraction	[1]
(ii)	Diffraction is a wave phenomenon electron can behave as a wave / have wave-like property	[1] [1]
(d)	• photon absorbed (by electron) and electron excited • photon energy equal to difference in (energy of two) discrete energy levels • photon energy relates to a single wavelength / single frequency • electron de-excites and emits photon in any direction (only some reach the screen)	[1] [1] [1] [1]
(e)	$\frac{hc}{\lambda} = \Delta E$ $\otimes E = E_3 - E_2$ $\frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{658 \times 10^{-9}} = E_3 - (-3.40 \times 1.6 \times 10^{-19})$ {As $E_3 - E_2$ is the lowest energy of all the transitions down to E_2 , thus $\otimes E = hc/\lambda$ corresponds to that of the longest wavelength} $E_1 = -2.42 \times 10^{-19} \text{ J}$	 [1 mk per eqn side] [1]

End of solutions